

Calibration

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 |
Generated: | Engine: DiscreteSim | Sections: 8

Calibration

The model parameters come from measurements, and every fit is tested against the observations it claims to describe.

The observations

Plant history

Series	N	Mean	Sd	Cv	Min	P50	P95	Max	Lag1
interarrival	54127	1.197	1.218	1.017	7.96e-6	0.825	3.595	15.1	0.019
service	48780	1.328	0.627	0.472	0.171	1.201	2.519	8.086	0.0012
failure_interval	354	172.3	113.9	0.661	3.02	148.8	387.4	643.5	-0.038
repair	354	11.1	6.141	0.553	2.991	9.591	22.4	42.3	-0.014
demand_size	82	14.4	8.029	0.556	4.547	12.9	23.2	50.3	-0.139

Days 45

Base rate 0.8

Service mean 1.2

Failure mtbf 180

Drift 0.25

Maintenance day 45

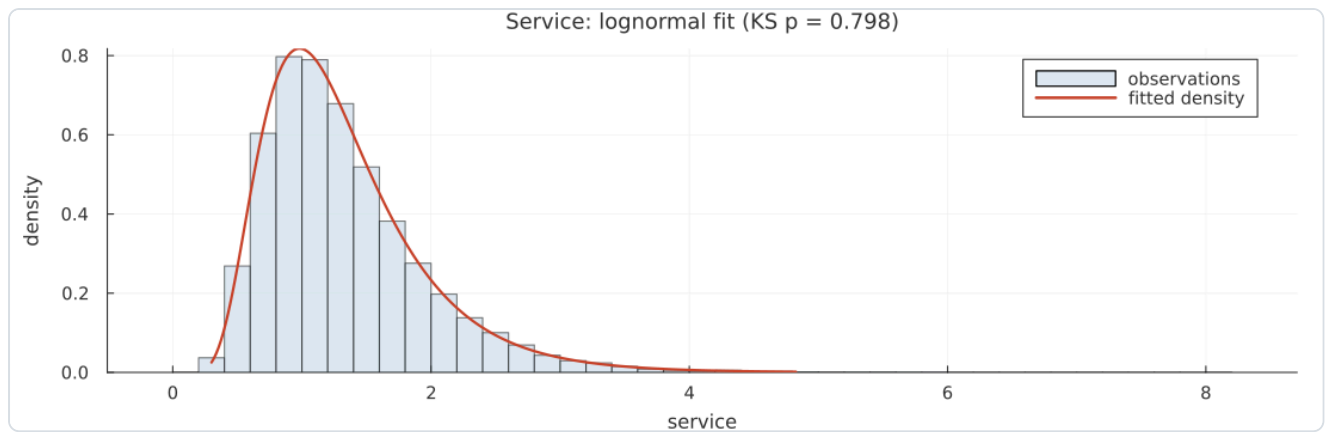
Service improvement 0.12

Seed 20260101

Maximum-likelihood fits

Fits

Series	Family	Mean	Sample mean	Cv	N	Ks stat	Ks p	Verdict
interarrival	gamma	1.197	1.197	1.005	54127	0.00293	0.741	validated
service	lognormal	1.329	1.328	0.473	48780	0.00292	0.798	validated
failure_interval	weibull	172.3	172.3	0.663	354	0.024	0.988	validated
repair	lognormal	11.1	11.1	0.529	354	0.044	0.488	validated
demand_size	lognormal	14.3	14.4	0.488	82	0.056	0.957	validated



Calibration

What the data implies for the model

Arrival rate 0.835

Service rate 0.753

Mtbf 172.3

Mttr 11.1

Availability 0.94

Demand mean 14.3